

PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	Make 2-3 notes of specific business requirements that will be analyzed. • <i>Will the app process transactions?</i> • <i>Does it do a lot of back-end processing?</i> • <i>Are there industry regulations that need to be considered?</i> Users need to be able to sign in, login, contact sellers and make purchases using payment options . The company must be able to store all the users credentials as well as their payment options in a secured database. Relevant SPII and PII protection laws/ regulations must be considered.
II. Define the technical scope	List of technologies used by the application: • <i>Application programming interface (API)</i> • <i>Public key infrastructure (PKI)</i> • <i>SHA-256</i> • <i>SQL</i> Write 2-3 sentences (40-60 words) that describe why you choose to prioritize that technology over the others. We can use SHA-256 in order to hash the users credential for them to be stored in a secured and unreadable manner. SQL can be used to store all the users data in a structured way and also improve the product search process in the database;
III. Decompose application	Sample data flow diagram
IV. Threat analysis	List 2 types of threats in the PASTA worksheet that are risks to the information being handled by the application. • <i>What are the internal threats?</i> • <i>What are the external threats?</i>

	Possible threat are SQLi and lack of prepared statement.
V. Vulnerability analysis	List 2 vulnerabilities in the PASTA worksheet that could be exploited. • <i>Could there be things wrong with the codebase?</i> • <i>Could there be weaknesses in the database?</i> • <i>Could there be flaws in the network?</i> • The product search process could be a vulnerability that can be exploited by SQLi if we do not have prepared statements. The apps authentication system could be attacked with a virus. Authentication could also be attacked if a threat actor social engineers an employee.
VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	List 4 security controls that you've learned about that can reduce risk. First, we can implement MFA for every user / seller to prevent credentials from being exploited . We also have to sanitise the users input before being sent to the database. For session hijacking , we can use temporary sessions cookies that last only for a period of time so that each session has a new cookie that expires. For weak login credentials, we can make strong passwords required , having more characters and specialchars.
